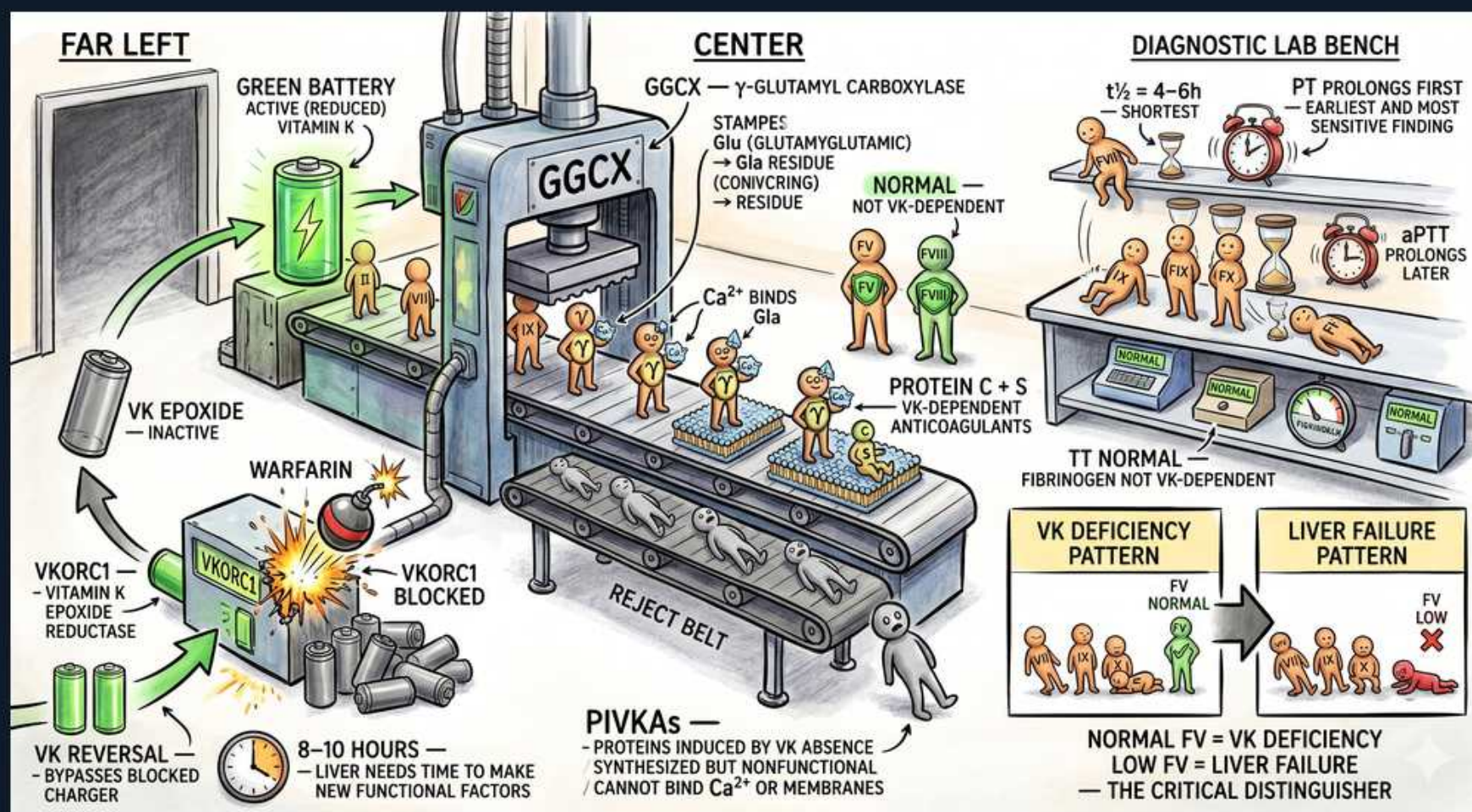


Coagulation — Vitamin K Deficiency (mechanism & labs)



1. Reduced vitamin K is active

WHAT YOU SEE

A glowing green BATTERY at the far left labelled 'ACTIVE (REDUCED) VITAMIN K' — the charged battery IS the active reduced (hydroquinone) cofactor; spend it and it becomes the dead, empty epoxide.

WHAT IT ENCODES

Vitamin K cycles between two forms: the REDUCED (hydroquinone) form is the ACTIVE cofactor, and it is oxidized to the inactive vitamin K EPOXIDE during each carboxylation reaction. The enzyme VKORC1 then regenerates the reduced form so the cycle can continue. Mnemonic: 'reduced = ready (charged battery); epoxide = empty.'

BOARD TRAP

The oxidized vitamin K EPOXIDE is INACTIVE until VKORC1 recycles it back to the reduced form — and VKORC1 is exactly where warfarin acts.

2. GGCX gamma-carboxylase

WHAT YOU SEE

The big central PRESS MACHINE stamped 'GGCX' pressing factors on a conveyor belt — the stamping press literally IS the carboxylase that hammers a -COOH onto each Glu residue (Glu → Gla).

WHAT IT ENCODES

Gamma-glutamyl carboxylase (GGCX) is the vitamin K–dependent enzyme that adds a carboxyl (-COOH) group to specific glutamate (Glu) residues, converting them to gamma-carboxyglutamate (Gla) on factors II, VII, IX, X and proteins C/S. This post-translational modification is what makes the factors functional. Mnemonic: 'GGCX is the press that stamps Glu → Gla.'

BOARD TRAP

Without carboxylation the factor PROTEINS are still synthesized and secreted — they are just NONFUNCTIONAL (these are the PIVKAs). So factor antigen can be normal while activity is low.

3. Factors II, VII, IX, X

WHAT YOU SEE

Little carboxylated SOLDIERS labelled IX, X, VII, II marching off the GGCX press onto the conveyor belt — the soldiers coming off the line are the vitamin K-dependent procoagulant factors, freshly stamped and functional.

WHAT IT ENCODES

The vitamin K–dependent PROCOAGULANTS are factors II (prothrombin), VII, IX, and X — all require gamma-carboxylation so their Gla domains can chelate Ca^{2+} and dock onto phospholipid membranes for cascade assembly. Mnemonic: '1972 — the years vitamin K depends on (factors 10, 9, 7, 2).'

BOARD TRAP

Factor V is NOT vitamin K–dependent (it's made by the liver but needs no carboxylation) — this is the key fact that separates vitamin K deficiency (FV normal) from liver failure (FV low).

4. Protein C & S also VK-dependent

WHAT YOU SEE

Center-right 'Protein C + S, VK-dependent anticoagulants'

WHAT IT ENCODES

The natural ANTICOAGULANTS protein C and protein S are ALSO vitamin K–dependent. Because protein C has a SHORT half-life (~8 h, similar to factor VII), it is depleted FIRST when warfarin starts — transiently before the procoagulants fall — creating an early hypercoagulable window. Mnemonic: 'C falls first → clots first.'

BOARD TRAP

Warfarin's transient drop in protein C is the mechanism of warfarin-induced SKIN NECROSIS (and is worsened in underlying protein C deficiency). This is why warfarin is BRIDGED with heparin at initiation and never started alone in acute thrombosis.

5. Carboxylation needs Ca^{2+}

WHAT YOU SEE

The ' Ca^{2+} BINDS Gla' note over a factor on the belt — the calcium ion clipping onto the new Gla residue is what lets the carboxylated factor grab the phospholipid membrane.

WHAT IT ENCODES

The gamma-carboxyglutamate (Gla) residues created by GGCX form negatively charged pockets that bind Ca^{2+} , which in turn tethers the factor to negatively charged phospholipid membranes (e.g., activated platelet surface). This membrane localization is essential for tenase and prothrombinase complex assembly. Mnemonic: 'Gla grabs calcium, calcium glues to membrane.'

BOARD TRAP

UNcarboxylated factors cannot bind Ca^{2+} or membranes — these des-gamma-carboxy proteins (PIVKAs) circulate but cannot participate in the cascade.

6. Warfarin blocks VKORC1

WHAT YOU SEE

Bottom-left bomb 'warfarin → VKORC1 blocked'

WHAT IT ENCODES

Warfarin (a vitamin K antagonist) inhibits VKORC1 (vitamin K epoxide reductase complex 1), blocking regeneration of reduced vitamin K. With no recycled active cofactor, GGCX cannot carboxylate new factors → functional vitamin K deficiency. Mnemonic: 'warfarin bombs the recycler.'

BOARD TRAP

Warfarin does NOT destroy factors already in circulation — it only prevents synthesis of new FUNCTIONAL ones. That's why the anticoagulant effect is DELAYED (~2–3 days, until existing factors decay) rather than immediate.

7. Vitamin K reversal bypasses block

WHAT YOU SEE

Bottom-left 'VK reversal bypasses blocked charger'

WHAT IT ENCODES

Administering vitamin K replenishes the REDUCED form directly, BYPASSING the VKORC1 block so GGCX can resume carboxylation. This is the rationale for vitamin K reversal of warfarin / treatment of deficiency. Mnemonic: 'flood the system with charged batteries to bypass the broken charger.'

BOARD TRAP

Vitamin K reversal is SLOW — it takes ~8–10 h (often 12–24 h for full effect) to synthesize new functional factors. For EMERGENT reversal you must add 4F-PCC (or FFP) for immediate factor replacement; vitamin K alone is not enough for life-threatening bleeds.

8. PIVKAs — nonfunctional factors

WHAT YOU SEE

Bottom-center 'PIVKAs — proteins induced by VK absence, cannot bind Ca^{2+} '

WHAT IT ENCODES

PIVKAs (Proteins Induced by Vitamin K Absence/Antagonism) are the des-gamma-carboxy factors produced when vitamin K is deficient or blocked — present in plasma but unable to bind Ca^{2+} /membranes, so functionally inert. (PIVKA-II / des-gamma-carboxy prothrombin is also a tumor marker in hepatocellular carcinoma.) Mnemonic: 'PIVKA = Proteins In a Vitamin-K-less state.'

BOARD TRAP

Factor antigen/protein may be PRESENT and normal by immunoassay while factor ACTIVITY is low — measuring antigen instead of clotting activity will miss the defect.

9. PT prolongs first

WHAT YOU SEE

Diagnostic bench: 'PT prolongs first, earliest most sensitive'

WHAT IT ENCODES

PT/INR rises FIRST in vitamin K deficiency and on starting warfarin because it depends on factor VII, which has the SHORTEST half-life (~4–6 h) of all the factors — so VII falls fastest. PT is therefore the earliest, most sensitive screen. Mnemonic: 'Seven is the shortest → PT moves first.'

BOARD TRAP

It is PT (not aPTT) that is the earliest and most sensitive marker. Expecting the aPTT to move first is the classic distractor.

10. aPTT prolongs later

WHAT YOU SEE

'aPTT prolongs later' hourglass

WHAT IT ENCODES

As deficiency persists, the longer-half-life factors II, IX, and X also fall, so the aPTT prolongs LATER; eventually BOTH PT and aPTT are abnormal. Sequence: PT first → then aPTT → then both. Mnemonic: 'VII leaves first (PT), the rest follow (aPTT).'

BOARD TRAP

A NORMAL aPTT early does NOT exclude vitamin K deficiency — only the PT may be prolonged at first. Don't rule it out on a normal aPTT.

11. TT stays normal

WHAT YOU SEE

'TT normal, fibrinogen not VK-dependent'

WHAT IT ENCODES

The thrombin time (TT) stays NORMAL in vitamin K deficiency because fibrinogen (factor I) and the thrombin-to-fibrin step are NOT vitamin K–dependent. A normal TT with prolonged PT/aPTT helps point to a factor-synthesis/carboxylation problem rather than a fibrinogen problem. Mnemonic: 'K doesn't touch fibrinogen, so TT is fine.'

BOARD TRAP

An ABNORMAL TT points AWAY from pure vitamin K deficiency — toward DIC, dysfibrinogenemia (liver disease), heparin contamination, or hypofibrinogenemia.

12. Factor V normal in VK def

WHAT YOU SEE

Right box 'VK deficiency pattern: FV normal'

WHAT IT ENCODES

Factor V is NORMAL in vitamin K deficiency because FV is not vitamin K–dependent. So the deficiency pattern is: low II/VII/IX/X (and protein C/S) but NORMAL factor V and normal fibrinogen. This normal FV is the key fingerprint. Mnemonic: 'V is fine when it's just K.'

BOARD TRAP

A LOW factor V means the problem is NOT isolated vitamin K deficiency — it indicates liver synthetic failure (where FV also drops). Use FV to triage between the two.

13. Liver failure: FV also low

WHAT YOU SEE

Right box 'liver failure pattern: FV low'

WHAT IT ENCODES

In LIVER failure factor V (non-VK-dependent, hepatically synthesized) is ALSO low — the critical lab that distinguishes hepatic synthetic failure from pure vitamin K deficiency. Pairing: low FV + low FVIII → think DIC; low FV + normal/high FVIII → think liver; normal FV → vitamin K deficiency. Mnemonic: 'Five low = liver; Five fine = vitamin K.'

BOARD TRAP

Don't blame liver disease when factor V is NORMAL — that's vitamin K deficiency, which responds to vitamin K. Conversely a low FV won't correct with vitamin K.